

ARAS: LLM-Supported Augmented Reality Assistance System for Pancreatic Surgery

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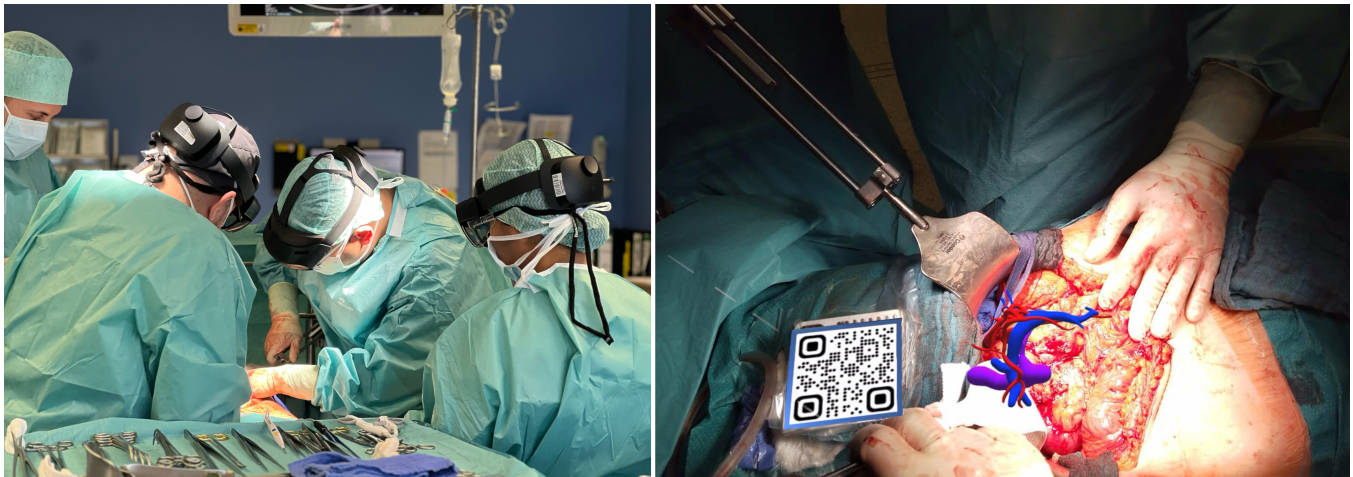


Figure 1: Pictures captured during clinical trials demonstrating the third and first-person view of surgeons using ARAS during an actual pancreatic surgery session

ABSTRACT

The integration of Augmented Reality (AR) technology into surgical procedures offers significant potential to enhance clinical outcomes. Despite numerous lab-proven prototypes, deploying these systems in actual clinical settings demands specialized design and rigorous clinical evaluation to meet the high standards of complex medical fields.

*Both authors contributed equally to this research.

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Our research highlights the intricate requirements emerging from clinical environments, particularly operating theaters. To address these challenges, we introduce ARAS, an operational AR assistance system for open pancreatic surgery. Through a user-centric design methodology, ARAS was iteratively developed and refined, ensuring its practical applicability and effectiveness in real-world surgical settings during clinical trials.

ARAS encompasses two different modes designed for preoperative and intraoperative sessions. Preoperative mode enables surgeons to visualize patient data and perform resection on the patient's 3D reconstructed vessels and tumors, helping surgeons during surgical planning. Intraoperative mode allows in-situ and precise visualization of these reconstructed 3D models during the surgical procedure and supports surgeons in decision-making.

ARAS also includes a novel interaction design for surgical AR systems. It uses LLMs to enable context-aware, intuitive, and natural communication between surgeons and the AR system. This

demo showcases the capabilities of ARAS, emphasizing its potential to transform surgical practices through advanced AR and LLM-supported interactions.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Mixed / augmented reality**; • **Applied computing** → **Life and medical sciences**.

KEYWORDS

Augmented Reality; Surgical Assistance System; Pancreatic Surgery; LLM

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1 INTRODUCTION

Augmented Reality (AR) assistance systems offer significant potential for surgical interventions, aiding surgeons by identifying vital structures, delineating dissection planes and margins, and minimizing damage to invisible structures or in cases of anatomical variations. However, many state-of-the-art methods are typically validated only in controlled lab environments and are often prone to underperform in actual surgical settings. The accuracy and validity of these methods are seldom tested under the unpredictable conditions encountered during live surgeries. Although many aspects of surgical rooms and surgical operations can be simulated in a lab, scenarios involving actual patients often cannot be fully replicated. Consequently, various interaction methods that could be easily employable in a simulated lab environment, face substantial challenges in real surgical contexts. Issues such as memorization problems due to extreme stress, time pressure, active bleeding, the crowded and noisy surgical scene, and unexpected changes during surgeries often cannot be fully simulated. These factors can significantly impact user interaction and the accuracy of the system. Even though finding general solutions that could be used in any context holds extreme importance in a wide range use of technologies such as AR, the systems and instruments designed for these special areas require specialized solutions that are tested in actual environments.

With latest development and the outburst of large language models (LLMs), many new opportunities have arisen. The usage of LLMs has shown great potential in different fields of the health industry [4, 18, 20, 29, 35]. While AR technology gathered great attention especially in surgical fields [6, 8, 9, 28], the integration of LLM in the medical domain are often targeting creation of chatbots and conversational assistants. The emergence of these two technologies beyond chatbots however is less explored.

While the AR navigational systems present great opportunities to increase surgical precision by providing assistance and accurate visualization of patient-specific 3D models during the operation, these systems often fall short due to the difficulties with interaction

during a highly stressful and critical environment [5, 10, 28]. The usage of LLM could help to decrease the cognitive workload caused by interacting with such systems by maintaining a natural communication scheme, hence promoting their usage in such critical domains.

Conventional methods of input, such as hand gestures or combinations of eye movements and virtual menus, are often not feasible where manual intervention is limited, and an uncluttered AR display is crucial, thus impeding the comprehensive adoption and effectiveness of wearable AR in these vital areas [24, 33].

In this work, we present ARAS, our AR-based surgical assistance system for pancreatic surgery encompassing planning and intraoperative modes [16, 34]. We will demonstrate the unique features of our system including resection simulation for surgical planning and a novel user interface using voice input in conjunction with LLMs, aiming to address the aforementioned limitations and support surgeons before and during surgical interventions.

2 RELATED WORK

2.1 AR in Surgical Domain

The field of surgical assistance and simulation has seen notable improvements through the adoption of AR technologies [22, 32, 36]. Numerous techniques have been applied in display-based or hand-held AR navigation systems, particularly in procedures like laparoscopic surgery [27]. However, the emergence of wearable AR devices has catalyzed the potential for deeper technological integration in more complex arenas such as open surgery [9, 26, 30]. While traditional interaction mechanisms using displays and touch-sensitive devices like tablets are well-established, the development of interaction methods for wearable AR remains incomplete. In critical areas such as open surgery, where interaction options are limited, the full integration of this technology into routine practices poses significant challenges due to the high stakes and the potential distractions caused by technological devices. Research by Saito et al. [28] highlighted that the use of 3D holograms and manual manipulation in such settings increases cognitive load, requiring more physical effort and attention compared to traditional 2D supports.

2.2 Interaction Techniques in the Medical Domain

The interaction techniques utilized in wearable AR technologies can differ based on the task and the modalities employed [12]. While a variety of interaction modalities are applicable in less critical medical fields, such as training and simulations [3, 17], the choices are considerably limited in sterile surgical settings. In such environments, hand, voice, and foot inputs have emerged as viable options since they do not involve direct contact with external objects [11]. However, the practical application of these interaction methods within the constraints of a surgical theater can be challenging [24, 33]. Furthermore, the limited range of gestures possible with hand or foot inputs may constrain the functionality of the system compared to voice-driven inputs [12].

2.3 Natural Communication and Speech Based Interaction

The utilization of voice input as a natural interaction and communication method has received considerable attention in research. Historically, most applications combined voice with other modalities [14, 19, 25], but advancements in speech recognition algorithms and natural language processing have enabled voice-based interaction as the sole input modality [31]. With the advent of smart assistant technologies and natural communication protocols, voice input has gained popularity, particularly in scenarios where other modalities are impractical [13]. The development of LLMs has further expanded the potential for natural communication in interaction with assistant systems. Mahmood et al. [23] introduced an LLM-powered conversational voice assistant applicable across various domains. Additionally, the integration of speech inputs with LLMs has proven effective in enhancing assistant system functionalities, as shown by Dong et al. [7].

3 SYSTEM DESIGN AND IMPLEMENTATION

The Surgical AR assistance system design process followed an iterative user-centric approach, marked by close collaboration with surgeons to gain a comprehensive understanding of the specific requirements and needs of open pancreatic surgery. The assistance system provides a dual mode to use prior surgery as a surgical planning tool and also an intraoperative mode to provide in-situ visualization of patient-specific 3D reconstructed models during the surgical procedure. We used the Unity 3D game engine [2] to design and develop the system and deployed it on the Microsoft HoloLens 2 device [1]. The selection of the HoloLens 2 for this application was mainly motivated by ethical, and safety considerations as it is CE-certified and was shown to have been successfully used in the medical domain before [15, 21, 28].

3.1 ARAS Planning

ARAS planning allows the visualization of patient's vascular system and tumor in a real-life size 3D model. This is crucial as sometimes making the connection between 2D CT images and the patient's anatomy can be challenging, especially for junior surgeons. Furthermore, we designed a resection simulation within ARAS-planning that allows surgeons to perform resections on the 3D reconstructed model prior to surgery, aiding them in developing a surgical plan. For resection simulation, we adopted a dynamic mesh cutting approach with a scalable cutting plane attached to the tip of a virtual scalpel (Figure 2). Surgeons can move this scalpel to cut the target 3D structure at the desired point and angle, with the resection occurring upon collision between the plane and the target structure. A cut is made if the virtual scalpel maintains contact with the target structure for at least 2 seconds, dividing the target 3D object mesh into two separate pieces at the collision point

3.2 ARAS Intraoperative

The fundamental functionality of the intraoperative ARAS application was to accurately overlay patient-specific, virtually reconstructed 3D structures onto the patient's abdomen area during the operation. An additional feature that proved to be useful during surgery was the ability to visualize the patient's history and CT

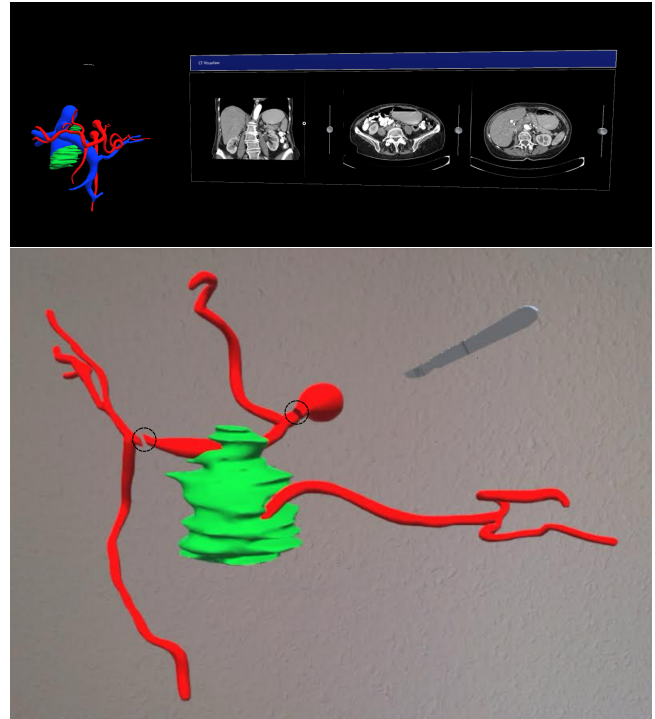


Figure 2: Data visualization and resection simulation features implemented in ARAS-Planning

images. Traditionally, observing CT images and being reminded of the patient's diagnosis and medications are routine practices during surgical procedures. This functionality proved to be a valuable addition, eliminating the need for an additional operation technician to scroll through CT images using a computer and monitor, a task surgeons cannot perform themselves due to sterilization rules.

To implement the essential functionality of the ARAS intraoperative system, we utilized a marker-based tracking approach. Our initial investigations, along with interviews with surgeons, indicated that the positional and rotational relationship between the sternum and vital structures (for pancreatic surgery) remains constant during the operation. Therefore, we selected the sternum as our anchor point for attaching our marker (Figure 3).

To overcome the interaction challenges caused by the spatial limitations around the surgical table, environmental conditions and the high stress level, we developed a voice-based interaction method to interact with our surgical AR assistance system to provide an optimal interaction method and reduce surgeons' cognitive workload. We used a novel method incorporating LLM and natural communication schemes enabling users to execute desired functionalities. We designed and implemented a system framework to automatically execute system functions upon user query stated via natural speech and voice input. We used the GPT-3.5-turbo model to process the user request and return the associated system functions.



Figure 3: The attached sterilized 3D printed marker on the patient sternum after identifying the anchor point done by a surgeon

4 EVALUATION

The entire process undertaken in this study received the approval of ethical committee board of Saarland medical association, ensuring that all aspects of the research adhered to established ethical guidelines. We evaluated our system in a highly ecologically valid environment through clinical trials during actual surgeries involving expert and specialized surgeons and further assessed the real-world impact of such assistance during surgical interventions (Figure 4). Over the series of these clinical trials, the system was evaluated by expert surgeons in terms of the accuracy of model overlay, overall effectiveness of AR support throughout the patient therapy pipeline [16, 34], and cognitive load assessment using LLM and voice-based natural communication as means of interaction.

5 DEMONSTRATION

In this demonstration, we will present unique features of our developed system ARAS, showcasing the dual-mode functionality to engage the audience and provide a better understanding of the challenging conditions during pancreatic surgery.

5.1 Preoperative Planning Mode

In this mode, attendees will witness the integration and visualization capabilities of ARAS in the context of surgical planning. Participants will explore a detailed 3D model reconstructed from patient-specific CT scans. This immersive experience will highlight how surgeons can manipulate and examine complex anatomical structures, perform resection simulations, and strategize surgical approaches effectively, ensuring optimal patient outcomes.

5.2 Intraoperative Mode

Transitioning from preoperative planning to the operating theater scenario, this part of the demonstration will focus on the intraoperative utility of ARAS. By using a 3D printed model of vessels and organs derived from an actual patient case, we will demonstrate the system's precision in overlaying critical anatomical information, such as tumor locations and vascular structures, directly into the surgeon's field of view. This real-time guidance is critical in navigating the unique challenges of each surgery, particularly in complex or extreme cases where conventional methods fall short.

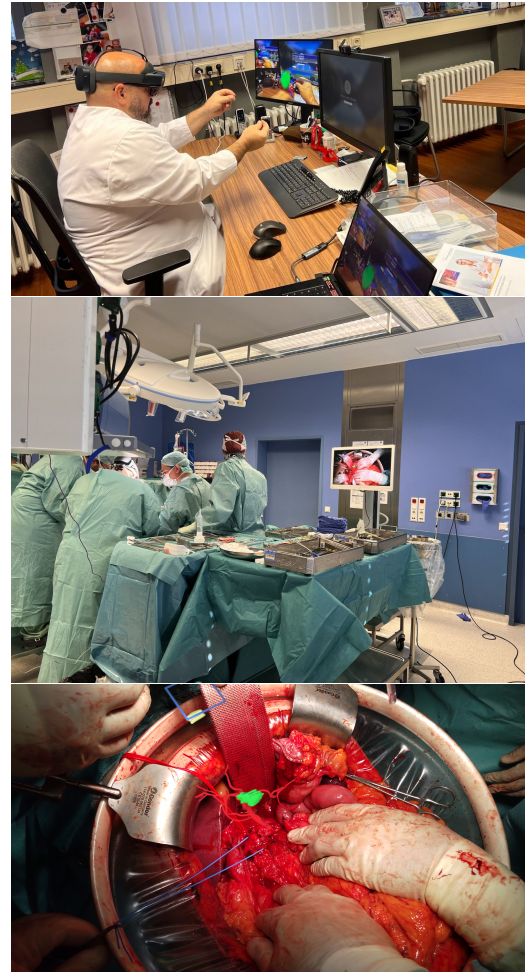


Figure 4: Pictures showing the usage of ARAS throughout a patient therapy pipeline, including planning and intraoperative sessions

Each demonstration will also highlight the system's user interaction, supported by llm that allows surgeons to query and retrieve specific information hands-free, further enhancing the surgical workflow. By engaging with ARAS through both preoperative and intraoperative modes, attendees will gain a profound understanding of how augmented reality can transform surgical planning and execution, leading to safer and more effective procedures.

6 CONCLUSION

In this work, we presented ARAS, an AR assistance system for pancreatic surgery. The system was designed to integrate into surgical workflows, offering both preoperative planning tools and intraoperative navigation assistance allowing in situ visualization of patient-specific 3D models of vascular systems and tumors. We validated our approach through clinical trials involving patients with pancreas cancer, demonstrating the feasibility of using ARAS for both preoperative and intraoperative sessions.

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